Lecture 7

Loss

linear regression - least squares

$$S = (X, y) \qquad (X \in \mathbb{R}^{n \times p}, y \in \mathbb{R}^n)$$

$$\begin{aligned} L_S(w) &= \frac{1}{2}\|Xw - y\|^2 \\ &= \frac{1}{2}(Xw - y)^\top (Xw - y) \\ &= \frac{1}{2}\left(w^\top X^\top X w - 2y^\top X w + y^\top y\right) \end{aligned}$$

- For a linear regression, the loss is quadratic, and thus *convex*. Therefore, it is easy to find a minimum.[54]
- However, for a DNN, the loss is generally *nonconvex*. So we need to use gradient-based optimization (e.g., GD) to find a minimum.

[54] Q. Find a minimum (the minimum?).

Q. Summarize the last lecture and compute $L_S(w)$ for a nonlinear basis model $h(x; w) = w^\top \phi(x)$ in the least squares setting (Is it quadratic?).

linear model $\rightarrow$ feature transformation (nonlinear basis function) $\rightarrow$ parameterized feature extractor $\rightarrow$ going deeper $\rightarrow$ inductive bias $\rightarrow$ less inductive bias

Preliminaries
- Tools
- Linear Algebra
- Matrix Calculus
- Probability

Introduction
- Neural Networks
- Optimization
- Neural Networks + Optimization
- Generalization
- etc.

Optimization

- population risk: $L_{\mathcal{P}}(h)$
- ERM

$$h_S \in \arg\min_{h \in \mathcal{H}} L_S(h)$$

- parameterized model h_θ
  - $L_S(\theta) := L_S(h_\theta)$
  - $\mathcal{H} = \{h_\theta : \theta \in \Theta\}$

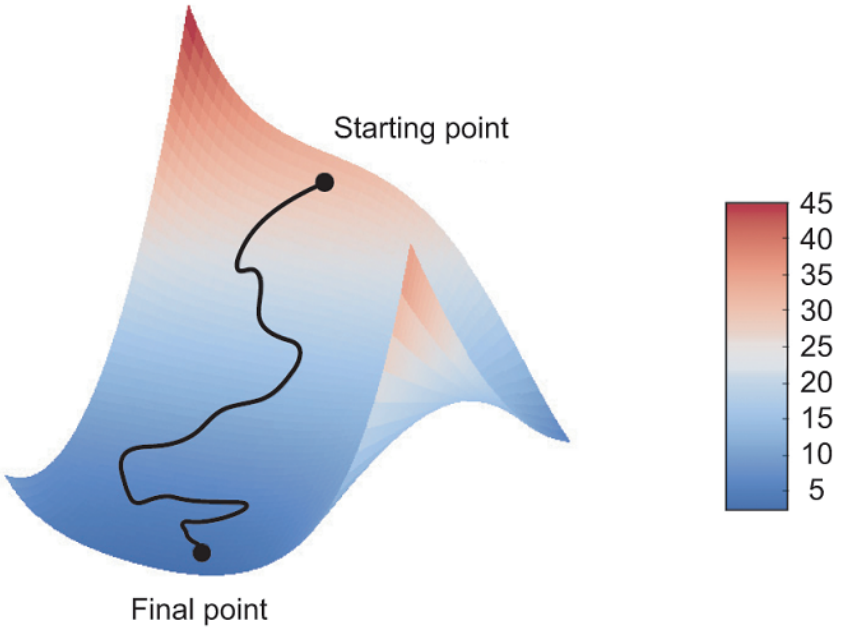
$$\theta_S \in \arg\min_{\theta \in \Theta} L_S(\theta) \qquad \text{(nonconvex)}$$

- GD; learning rate $\eta > 0$

$$\theta \leftarrow \theta - \eta \nabla_\theta L_S(\theta)$$

- SGD; mini-batch $B \sim \mathcal{P}^b$ ($|B| = b$)

$$\theta \leftarrow \theta - \eta \nabla_\theta L_B(\theta)$$


Starting point
Final point
45
40
35
30
25
20
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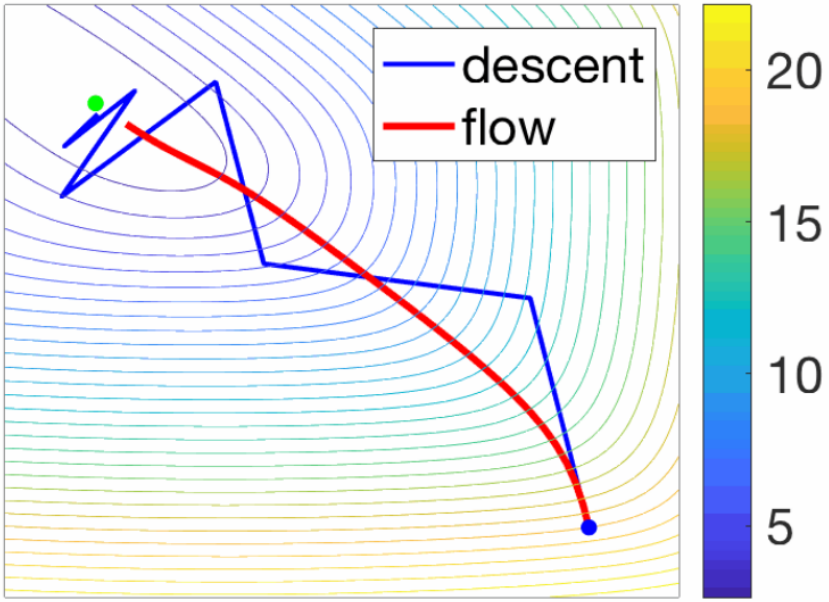
5


descent
flow
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5

A Pytorch Example

Taking an optimization step

All optimizers implement a `step()` method, that updates the parameters. It can be used in two ways:

`optimizer.step()`

This is a simplified version supported by most optimizers. The function can be called once the gradients are computed using e.g. `backward()`.

Example:

```
for input, target in dataset:
    optimizer.zero_grad()
    output = model(input)
    loss = loss_fn(output, target)
    loss.backward()
    optimizer.step()
```

Momentum Method

Now, we use m instead of $g(= \nabla_\theta L_B(\theta))$

$$\theta \leftarrow \theta - \eta m.$$

- heavy ball (momentum); $0 \leq \beta \leq 1$[55]

$$m \leftarrow \beta m + g(\theta)$$

- Nesterov accelerated gradient (NAG)[56]

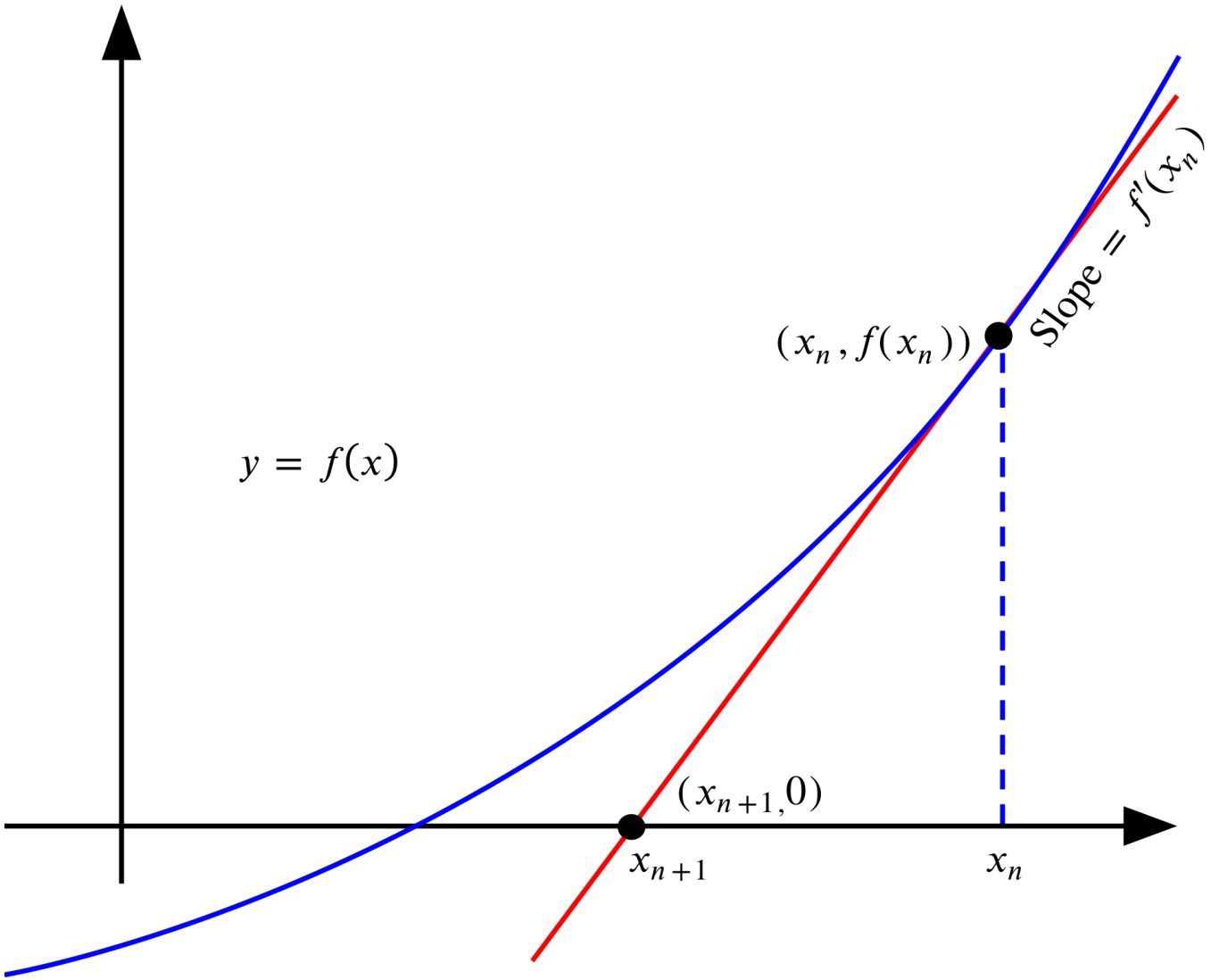
$$m \leftarrow \beta m + g(\theta - \beta m)$$

[55] How about the EMA $(\beta, 1-\beta)$-weighting? (Quiz)

[56] ... or equivalently, (Quiz)

$$\tilde{\theta}_t \leftarrow \theta_t + \beta(\theta_t - \theta_{t-1}) \qquad \text{(look ahead)}$$
$$\theta_{t+1} \leftarrow \tilde{\theta}_t - \eta g(\tilde{\theta}_t)$$

Q. Newton's Method
(to find zeros $f(x) = 0 \rightarrow$ to find optima $L'(\theta^*) = 0$)



$$L'(\theta^*) = f(\theta^*) = 0 \leftrightarrow \theta^* \in \arg\min_\theta L(\theta)$$

preconditioned GD/SGD

Now, we use $M^{-1}g$ instead of g

$$M = \mathrm{diag}(\sqrt{s+\varepsilon}) \qquad (s : \text{vector},\ \varepsilon : \text{scalar})$$
$$\theta \leftarrow \theta - \eta M^{-1} g(\theta).$$

- AdaGrad [DHS11]

$$s \leftarrow s + g^2 \qquad \left(s_t = \sum_{i=1}^{t} g_t^2\right)$$

- RMSProp [HSS12]

$$s \leftarrow \beta s + (1-\beta) g^2 \qquad (\sqrt{s} \approx \mathrm{RMS})$$

- Adam [KB14] (RMSProp + momentum); $0 \le \beta_1, \beta_2 \le 1$

$$m \leftarrow \beta_1 m + (1-\beta_1) g$$
$$s \leftarrow \beta_2 s + (1-\beta_2) g^2$$
$$\theta \leftarrow \theta - \eta M^{-1} m$$

Line Search

For a quadratic loss, ...

$$
\begin{aligned}
L(\theta) &= \frac{1}{2}\theta^\top A\theta + a^\top\theta + b && (A \succ 0; \text{pd}) \\
\delta &= -\eta d \\
\phi(\eta) &= L(\theta - \eta d) \\
&= L(\theta) - \eta g^\top d + \eta^2 \frac{1}{2} d^\top A d && (g = \nabla_\theta L(\theta)) \\
\left.\frac{\partial \phi(\eta)}{\partial \eta}\right|_{\eta=\eta^*} &= -g^\top d + 2\eta^* \frac{1}{2} d^\top A d = 0 \\
\eta^* &= \frac{g^\top d}{d^\top A d} && (\text{exact line search})
\end{aligned}
$$

Convergence

For a quadratic loss, ...

$$
\begin{aligned}
L(\theta) &= \frac{1}{2}\theta^\top A\theta + a^\top\theta + b && (A \succ 0; \text{pd}) \\
\delta &= -\eta g && (g = \nabla_\theta L(\theta)) \\
L(\theta+\delta) &= L(\theta) + \nabla_\theta L(\theta)^\top \delta + \frac{1}{2}\delta^\top \frac{\partial^2 L(\theta)}{\partial\theta^2}\delta \\
&\le L(\theta) - \eta\|g\|^2 + \frac{1}{2}\eta^2\lambda_{\max}(A)\|g\|^2 \\
&= L(\theta) - \eta\left(1 - \frac{1}{2}\lambda_{\max}(A)\eta\right)\|g\|^2
\end{aligned}
$$

Theorem

If we run GD on a quad. loss with $\eta < \frac{2}{\lambda_{\max}(A)}$ and $A = \frac{\partial^2 L}{\partial\theta^2} \succ 0$, then the loss decreases.[57]

[57] Actually, we have global convergence iff $\eta < \frac{2}{\lambda_{\max}(A)}$.

Convex Set

A set X is said to be convex set if

$$x, y \in X \rightarrow \lambda x + (1 - \lambda) y \in X, \ \forall \lambda \in [0, 1].$$

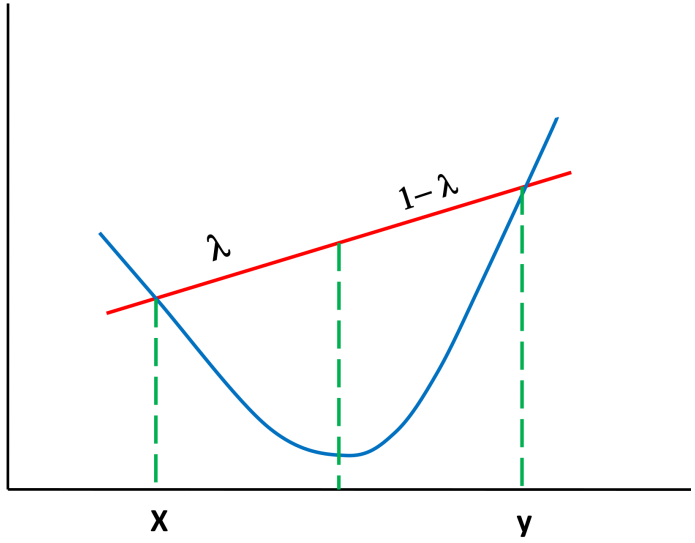
Convex Function

A function $f : X \to \mathbb{R}$ is said to be convex if X is convex and

$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y)$$

for any $x, y \in X$ and for any $\lambda \in [0, 1]$.

- convex ($\leq$), strictly convex ($<$), concave ($\geq$)
- A function g is concave iff $-g$ is convex.



- If f is differentiable and convex, then the following holds

$$f(x) \geq \underbrace{f(y) + \nabla f(y)(x-y)}_{\text{linear approx. at } y}$$

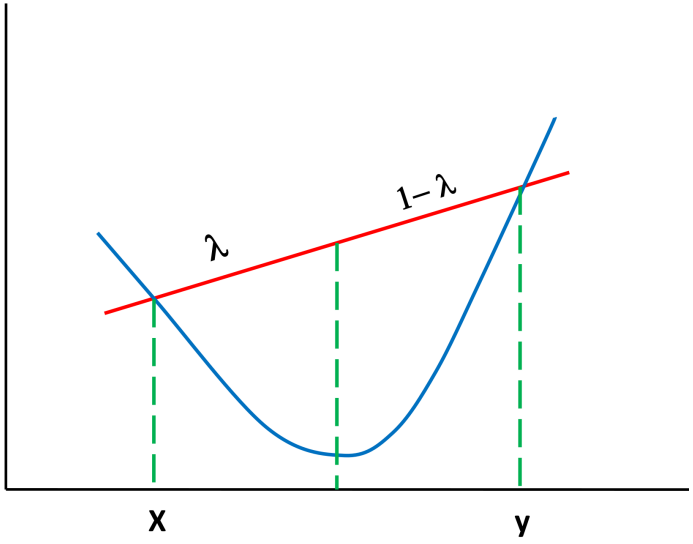
  for any $x, y \in X$.
- If f is differentiable, then f is convex iff, for any $x, y \in X$,

$$\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq 0$$

- If f is twice differentiable, then f is convex [strictly convex] iff

$$\nabla_x^2 f(x) \succeq 0 [\nabla_x^2 f(x) \succ 0]$$

  for any $x \in X$.



Theorem (Jensen's Inequality)

For a convex function $f : X \to \mathbb{R}$, we have

$$f(\sum_i \lambda_i x_i) \leq \sum_i \lambda_i f(x_i)$$

for any $x_i \in X$ and for any $\lambda_i \in [0, 1]$ such that $\sum_i \lambda_i = 1$. Similarly, we have[58]

$$f(\mathbb{E}_X[X]) \leq \mathbb{E}_X[f(X)]$$

[58] think of λ_i as a pdf $p_X(x)$.

Quiz

- Show that the KL-divergence is always nonnegative.
- A function $f(x) = x^\top Ax$ is convex if A is a psd matrix.

Convex

If f is differentiable, then f is convex iff, for any $x, y \in X$,

$$\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq 0.$$

(m-)Strongly Convex (not too flat)

A function $f : X \to \mathbb{R}$ is said to be m-strongly convex ($m > 0$) if

$$\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq m\|x - y\|^2$$

for any $x, y \in X$ where X is a normed space.

- strongly convex $\to$ strictly convex $\to$ convex (= "0-strongly convex")
- If f is twice differentiable, then f is strongly convex iff $\nabla_x^2 f(x) \succeq mI$

(m-)Strongly Convex (not too flat)

A function $f : X \to \mathbb{R}$ is said to be m-strongly convex ($m > 0$) if

$$\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq m\|x - y\|^2,$$

or equivalently, if

$$f(x) - \underbrace{\Big(f(y) + \nabla f(y)^\top (x - y)\Big)}_{\text{linear approx. at } y} \geq \frac{m}{2}\|x - y\|^2$$

for any $x, y \in X$ where X is a normed space.

- strongly convex $\to$ strictly convex $\to$ convex (="0-strongly convex")
- If f is twice differentiable, then f is strongly convex iff $\nabla_x^2 f(x) \succeq mI$.

Lipschitz Continuity (not too steep)

A function $f : X \to \mathbb{R}$ is L-Lipschitz continuous (L: Lipschitz constant) if, for any $x, y \in X$,

$$|f(x) - \underbrace{f(y)}_{\text{value at } y}| \leq L\|x - y\|.$$

β-smoothness (not too sharp)

A function $f : X \to \mathbb{R}$ is β-smooth for some constant $\beta > 0$ if

$$|f(x) - \underbrace{\Big(f(y) + \nabla f(y)^\top (x - y)\Big)}_{\text{linear approx. at } y}| \leq \frac{\beta}{2}\|x - y\|^2$$

for any $x, y \in X$ where X is a normed space.

- f is β-smooth iff ∇f is β-Lipschitz continuous.
- If f is twice differentiable, then f is β-smooth iff $\nabla_x^2 f(x) \preceq \beta I$

Polyak-Łojasiewicz condition (grad. not too small)

A function $f : X \to \mathbb{R}$ satisfies c-PŁ-condition if

$$\frac{1}{2}\|\nabla f(x)\|^2 \geq c(f(x) - f(x^*))$$

for any $x \in X$ where $f(x^*)$ is the optimal function value.

Linear Convergence

For a loss with β-smoothness (not too sharp) and c-PŁ-condition (grad. not too small), ...

$$
\begin{aligned}
L_{t+1} - L_t &\leq -\eta \|\nabla L\|^2 + \frac{\eta^2}{2}\beta\|\nabla L\|^2 && (\beta\text{-smoothness}) \\
&= -\eta\|\nabla L\|^2\left(1 - \frac{\eta\beta}{2}\right) \\
&= -\frac{1}{2\beta}\|\nabla L\|^2 && (\eta = \frac{1}{\beta}) \\
&\leq -\frac{c}{\beta}(L_t - L^*) && (c\text{-PŁ-condition})
\end{aligned}
$$

$$
L_{t+1} - L_t + (L_t - L^*) \leq (L_t - L^*) - \frac{c}{\beta}(L_t - L^*)
$$

$$
\begin{aligned}
L_{t+1} - L^* &\leq \left(1 - \frac{c}{\beta}\right)(L_t - L^*) \\
L_t - L^* &\leq \left(1 - \frac{c}{\beta}\right)^t (L_0 - L^*) && (L_t \in O(e^{-c't}))
\end{aligned}
$$

- second-order methods
- Newton's method
- quasi-Newton methods, (L-)BFGS
- trust-region methods
- learning rate schedule
- iterate averaging

SGD can often find surprisingly "good" solutions even though the DNN loss is not a convex objective.